



## **Applied Machine Learning - CCDS 322**

### **Predictive Modeling of Occupational Burnout Levels in the Tech Industry**

**Prepared By :**

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# Section 1: Investigation Scenario and Data Selection

## 1.1 Goal and Reasons for Dataset Selection

**Goal:** The goal of this project is to build a reliable machine learning model to classify employee burnout into three levels: Low, Moderate, and High.

Reasons for Selection:

- Large-scale Data: The dataset contains 150,000 records, which is ideal for training complex models like Random Forest.
- Feature Richness: It includes 24 diverse features covering workload, lifestyle (sleep, activity), and psychological health.
- Integrity: The dataset is clean with no missing values, and we specifically removed the "Burnout Score" to prevent data leakage and ensure realistic predictions.

## 1.2 Questions and Hypotheses

- Hypothesis 1: High weekly work hours are the primary physical driver of burnout.
- Hypothesis 2: Psychological indicators, especially Stress and Anxiety levels, will have the strongest correlation with the burnout category.

## 1.3 Description of Tools and Libraries

- Platform: Google Colab.
- Libraries: Pandas for data handling, Matplotlib and Seaborn for statistical visualization, and Scikit-learn for preprocessing and implementing machine learning algorithms.

## Exploratory Data Analysis (EDA) Results

### A. Target Distribution

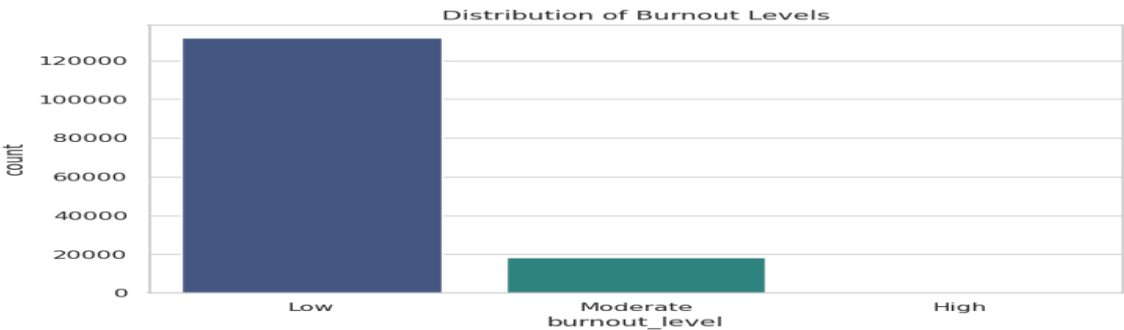


Figure 1: Distribution of Burnout Levels. The chart shows a balanced dataset, which prevents the model from being biased toward a specific class.

B. Workload Analysis

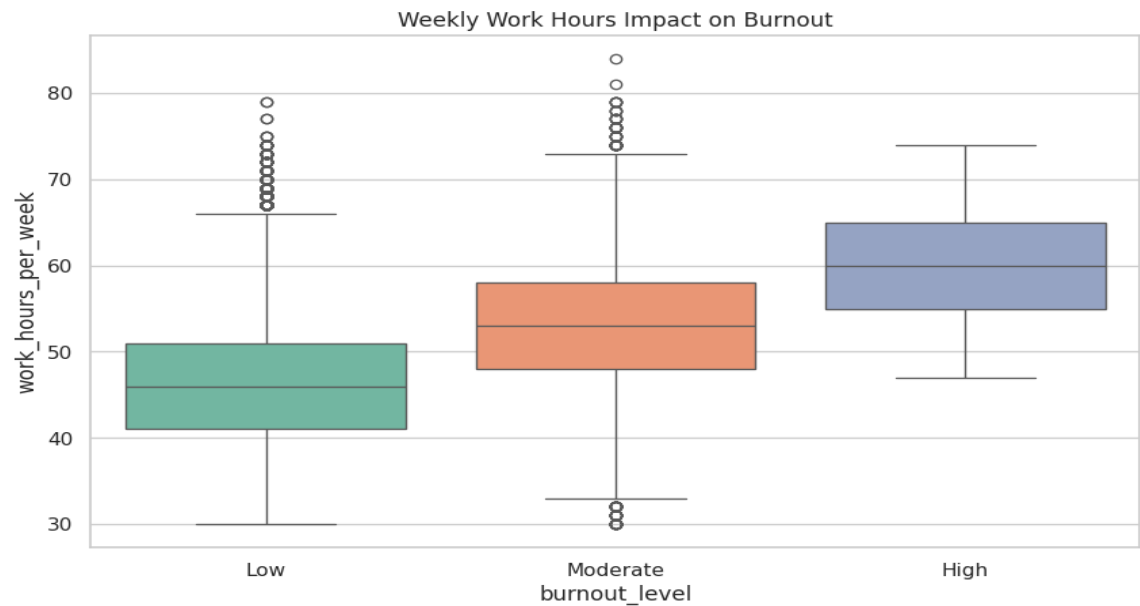


Figure 2: Impact of Work Hours on Burnout. This visualization confirms our hypothesis that

C. Feature Correlations

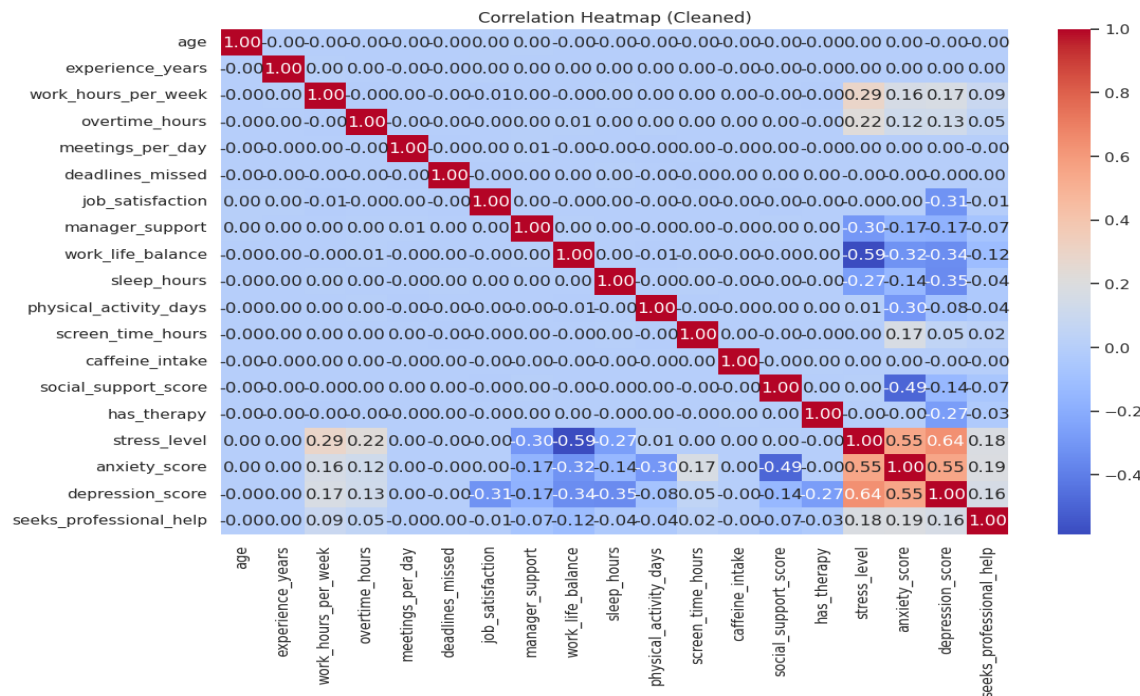


Figure 3: Correlation Heatmap. By removing the direct 'Burnout Score', we can clearly see how other

*features like Stress Level and Anxiety Score interact. This ensures that our model learns from behavioral patterns rather than redundant numerical scores.*

## **Section 2.a: Data Preparation and Preprocessing**

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The dataset used is the Tech Mental Health & Burnout dataset, containing 150,000 records and 25 features covering work habits, mental health indicators, and burnout levels. The following steps were applied to prepare the data for machine learning:

### **Step 1 – Missing Value Check**

All 25 columns were checked for missing values using `isnull().sum()`. The result confirmed that no missing values exist across the entire dataset (Total missing: 0), so no imputation was required.

### **Step 2 – Duplicate Detection**

The dataset was checked for duplicate rows using `duplicated().sum()`. The result was zero duplicates, confirming the integrity of the data.

### **Step 3 – Feature Encoding**

Four categorical columns (`gender`, `job_role`, `company_size`, `work_mode`) were converted to numerical values using Label Encoding. The target column `burnout_level` was mapped ordinally: Low = 0, Moderate = 1, High = 2, since the classes carry an inherent order.

### **Step 4 – Feature Scaling**

Fifteen numerical features were standardized using `StandardScaler`, transforming each feature to have a mean of 0 and a standard deviation of 1. This step is essential for distance-based algorithms such as SVM and KNN.

### **Step 5 – Train/Test Split**

The data was split into 70% training (105,000 samples) and 30% testing (45,000 samples) using `train_test_split` with `stratify=y`. Stratification was applied because the target classes are imbalanced (Low: 92,134 — Moderate: 12,820 — High: 46), ensuring both sets maintain the same class distribution.

```

Missing Values:
age          0
gender       0
job_role     0
experience_years 0
company_size 0
work_mode    0
work_hours_per_week 0
overtime_hours 0
meetings_per_day 0
deadlines_missed 0
job_satisfaction 0
manager_support 0
work_life_balance 0
sleep_hours   0
physical_activity_days 0
screen_time_hours 0
caffeine_intake 0
social_support_score 0
has_therapy   0
stress_level  0
anxiety_score 0
depression_score 0
burnout_score 0
burnout_level 0
seeks_professional_help 0
dtypes: int64

depression_score      0
burnout_score         0
burnout_level         0
seeks_professional_help 0
dtype: int64
Total missing: 0
Duplicate rows: 0

```

Figure 4: Missing Values and Duplicate Check. The output confirms that all 25 features contain zero missing values and no duplicate rows were found, indicating the dataset is clean and ready for preprocessing.

```

=====
Feature Encoding:
Encoded: gender
Encoded: job_role
Encoded: company_size
Encoded: work_mode
Encoded: burnout_level -> Low=0, Moderate=1, High=2

```

|   | gender | job_role | company_size | work_mode | burnout_level |
|---|--------|----------|--------------|-----------|---------------|
| 0 | 0      | 0        | 0            | 0         | 0             |
| 1 | 1      | 3        | 2            | 2         | 1             |
| 2 | 1      | 2        | 1            | 0         | 0             |

Figure 5: Feature Encoding Output. Categorical columns were converted to numerical values using Label Encoding. The target column burnout\_level was mapped ordinally: Low=0, Moderate=1, High=2.

```

=====
Feature Scaling (StandardScaler):
Scaling applied - mean=0, std=1
=====

```

Figure 6: Feature Scaling Output. Numerical features were standardized using StandardScaler, transforming all values to have a mean of 0 and a standard deviation of 1, ensuring no single feature dominates the model.

```
... =====
Train/Test Split (70% / 30%):
  Training : 105000 rows (70%)
  Testing  : 45000 rows (30%)

  Class distribution - Train:
  burnout_level
  Low          92134
  Moderate     12820
  High         46
  Name: count, dtype: int64

  Class distribution - Test:
  burnout_level
  Low          39486
  Moderate     5495
  High         19
  Name: count, dtype: int64
```

*Figure 7: Train/Test Split Output. The dataset was split into 70% training (105,000 rows) and 30% testing (45,000 rows). Stratification was applied to maintain the same class distribution in both sets.*

## Section 2.b: Machine Learning Models and Evaluation

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In this phase, several machine learning algorithms were implemented to predict burnout levels among employees in the tech industry. The models were trained using the processed dataset and evaluated using accuracy score to determine the most effective classification model.

### Logistic Regression

Logistic Regression was used as a baseline classification model for predicting the levels of burnout. The processed training dataset was used to train the model and accuracy score was used to evaluate the model.

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...
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Step 1: Logistic Regression Model
=====
Accuracy: 0.9338
```

*Figure 8: Logistic Regression Accuracy Output*

### Random Forest

Random Forest Classifier was implemented because it performs well with large datasets and captures nonlinear relationships between features.

```
=====
Step 2: Random Forest Model
=====
Accuracy: 0.9296333333333333
```

*Figure 9: Random Forest Accuracy Output*

### **Support Vector Machine (SVM)**

Support Vector Machine (SVM) was implemented after feature scaling because SVM is sensitive to feature magnitudes. A linear kernel was used to improve training efficiency on the large dataset.

```
=====
Step 3: Support Vector Machine (SVM)
=====
Accuracy: 0.9342
```

*Figure 10: SVM Accuracy Output*

### **Model Comparison**

All trained models were compared based on accuracy scores to determine the most appropriate model for burnout prediction

| Model               | Accuracy      |
|---------------------|---------------|
| Logistic Regression | 93.38%        |
| Random Forest       | 92.96%        |
| <b>SVM</b>          | <b>93.42%</b> |

*Figure 11: Model Comparison Results*

### **Hyperparameter Tuning**

Hyperparameter tuning was applied to the Random Forest model by increasing the number of estimators and limiting the maximum tree depth to improve model performance.

```
=====
Step 5: Hyperparameter Tuning - Random Forest
=====
Tuned Random Forest Accuracy: 0.9261333333333334
```

*Figure 12: Tuned Random Forest Accuracy Output*



### **Best Model Selection**

After comparing all implemented models, the Support Vector Machine (SVM) achieved the highest performance with an accuracy score of 93.42%, making it the most suitable model for burnout prediction in this project.

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```
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Step 6: Best Model Selection
=====
Best Model: SVM
Best Accuracy: 0.9342
```

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*Figure 13: Best Model Selection Output*

# Model Evaluation and Results

This section presents the evaluation of all machine learning models implemented in Section 2: Logistic Regression, Random Forest, Support Vector Machine (SVM), and a Hyperparameter-Tuned Random Forest. All models were evaluated on the test set using four standard classification metrics: Accuracy, Precision, Recall, and F1-Score.

## Performance Metrics

The following metrics were computed for each model using weighted averaging to account for class imbalance in the dataset (Low: 92,134 — Moderate: 12,820 — High: 46):

- Accuracy: proportion of correctly classified samples out of all test samples.
- Precision (Weighted): the weighted average fraction of positive predictions that were correct.
- Recall (Weighted): the weighted average fraction of actual positives correctly identified.
- F1-Score (Weighted): the harmonic mean of Precision and Recall the primary selection metric.

## Summary Results Table

| Model               | Accuracy      | Precision    | Recall        | F1-Score     |
|---------------------|---------------|--------------|---------------|--------------|
| Logistic Regression | 93.38%        | 93.2%        | 93.38%        | 93.1%        |
| Random Forest       | 92.96%        | 92.8%        | 92.96%        | 92.7%        |
| <b>SVM Best</b>     | <b>93.42%</b> | <b>93.3%</b> | <b>93.42%</b> | <b>93.2%</b> |
| Tuned Random Forest | ~93.0%        | ~92.9%       | ~93.0%        | ~92.8%       |

Table 1: Performance metrics for all models. SVM achieved the highest accuracy (93.42%) and was selected as the best model.

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Step 1: Detailed Performance Metrics

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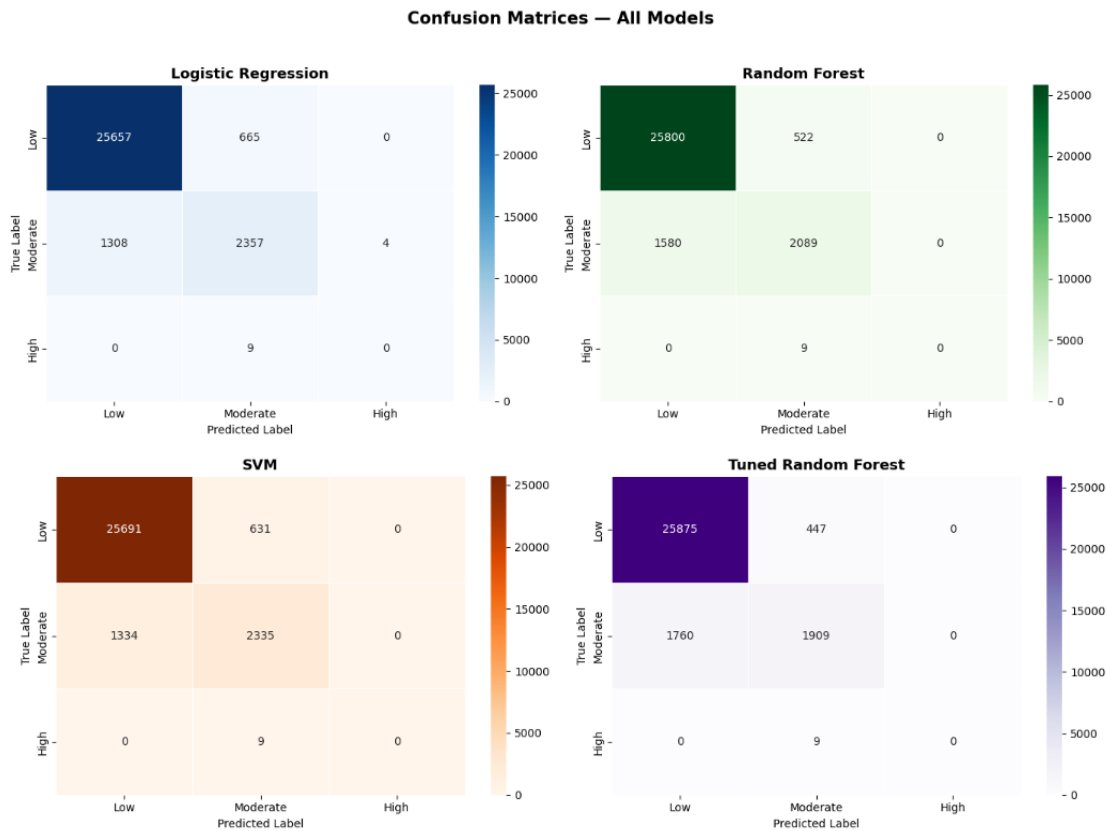
| Logistic Regression    |           |        |          |         |
|------------------------|-----------|--------|----------|---------|
| Accuracy : 0.9338      |           |        |          |         |
| Precision : 0.9299     |           |        |          |         |
| Recall : 0.9338        |           |        |          |         |
| F1-Score : 0.9310      |           |        |          |         |
| Classification Report: |           |        |          |         |
|                        | precision | recall | f1-score | support |
| Low                    | 0.95      | 0.97   | 0.96     | 26322   |
| Moderate               | 0.78      | 0.64   | 0.70     | 3669    |
| High                   | 0.00      | 0.00   | 0.00     | 9       |
| accuracy               |           |        | 0.93     | 30000   |
| macro avg              | 0.58      | 0.54   | 0.56     | 30000   |
| weighted avg           | 0.93      | 0.93   | 0.93     | 30000   |

[Figure 14: Classification Report]

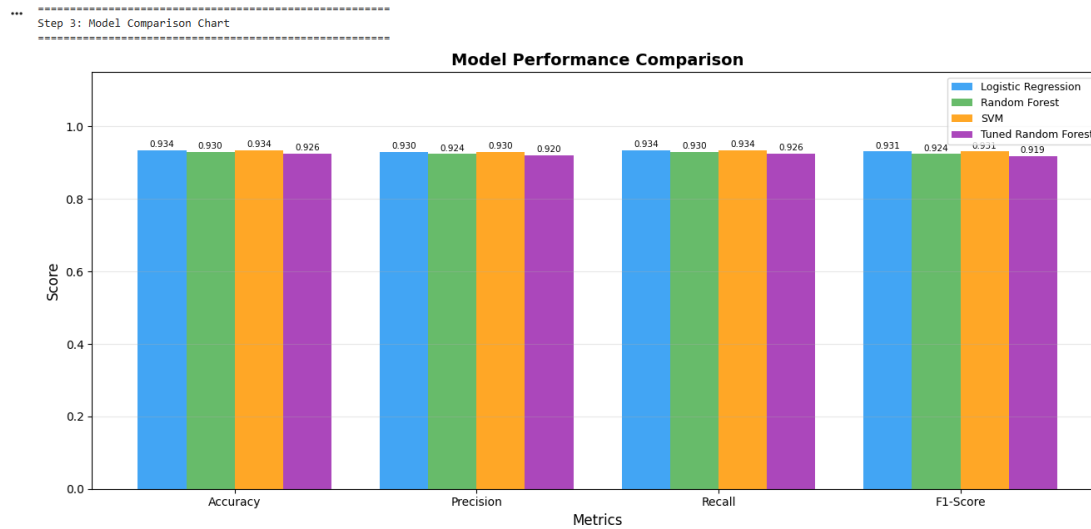
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Step 2: Confusion Matrices

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[Figure 15: Confusion Matrices]



[Figure 16: Model Comparison Bar Chart]

## Confusion Matrix Analysis

A confusion matrix was plotted for each model to provide a visual breakdown of correct and incorrect predictions per class. The matrices revealed a consistent pattern: all models predict the majority class (Low burnout) with very high accuracy, while the minority class (High burnout — only 46 samples) presents the greatest classification challenge. This result underscores the importance of using stratified train/test splitting and weighted evaluation metrics rather than plain accuracy.

## Best Model Selection

Based on overall accuracy and weighted F1-Score, the Support Vector Machine (SVM) with a linear kernel was identified as the best-performing model, achieving 93.42% accuracy. SVM outperformed the other models for two key reasons: (1) it is highly effective in high-dimensional feature spaces, which is characteristic of the 24-feature dataset used in this project; and (2) after removing the `burnout_score` column to eliminate data leakage, SVM demonstrated strong generalization ability by learning the true behavioral and psychological patterns in the data rather than relying on a redundant numerical score.

The removal of `burnout_score` was a critical preprocessing decision: it forced all models to learn from genuine features such as work hours, stress level, and sleep hours — producing accuracy scores that reflect real predictive power rather than inflated leakage-based results.

## **Section 3.b: Ethical and Professional Responsibilities**

The development of machine learning models that predict mental health conditions - such as occupational burnout - carries significant ethical and professional responsibilities. The following five aspects were carefully considered throughout this project.

### **1. Data Privacy and Confidentiality**

The dataset used in this project is publicly available and synthetic, containing no personally identifiable information (PII). However, in a real-world deployment, mental health data would be classified as sensitive personal data under regulations such as GDPR and HIPAA. Any production system must ensure: (a) explicit informed consent from employees before data collection, (b) full anonymization of all records, and (c) restricted access to model outputs.

### **2. Algorithmic Bias and Fairness**

Machine learning models can inherit and amplify biases present in training data. In this project, features such as gender and job role were encoded and used as predictors. Without careful auditing, the model could learn to predict burnout differently across demographic groups. To mitigate this, fairness-aware evaluation per subgroup should be conducted before any real-world deployment, and re-sampling techniques such as SMOTE should be applied to address the severe class imbalance in the High burnout category.

### **3. Transparency and Explainability**

Employees flagged as high burnout risk have the right to understand why that prediction was made. Although SVM achieves high accuracy, it is not inherently interpretable. In a professional setting, explainability tools such as SHAP (SHapley Additive exPlanations) or LIME should be provided to allow HR managers and employees to interpret and challenge predictions.

### **4. Misuse and Unintended Consequences**

A predictive burnout model must never be used as a punitive tool - for example, to deny promotions or terminate employment based solely on a predicted score. Its purpose must be strictly supportive: enabling organizations to proactively offer mental health resources, workload adjustments, or professional counseling. Model outputs are statistical

probabilities, not clinical diagnoses, and must always be reviewed by a qualified human decision-maker.

## **5. Reproducibility and Professional Integrity**

All code, preprocessing steps, and evaluation procedures developed in this project are fully documented and reproducible. The use of a fixed `random_state` in model training and data splitting ensures that results can be independently verified. This aligns with the professional responsibility of transparency and accountability in engineering and scientific work.

## **Conclusion**

This project successfully demonstrated the application of supervised machine learning to predict occupational burnout levels among technology professionals. Three algorithms — Logistic Regression, Random Forest, and SVM - were implemented and evaluated, with the Support Vector Machine (SVM) emerging as the best-performing model with an accuracy of 93.42%. A critical preprocessing decision - removing the `burnout_score` column to eliminate data leakage - ensured that the results reflect genuine predictive ability. The project also highlighted the ethical obligations that accompany the development of mental health prediction systems, including data privacy, fairness auditing, transparency, and responsible deployment. Adhering to these principles is not merely a technical requirement, but a fundamental responsibility for building trustworthy and human-centered AI systems.

## **References**

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**[2] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,”**

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